Euclid's Elements — Book I
Proposition 42

Formal Reconstruction — Technical Documentation

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CGJteamLab

Constructing a parallelogram equal in area to a given triangle

1 Introduction

Proposition I.42 is the first construction in Book I whose target is not merely a line, angle, triangle, or parallel relation, but a new figure with prescribed *content*. Given a triangle and a rectilineal angle, Euclid constructs a parallelogram equal to the triangle and having an angle equal to the prescribed one. In the traditional sequence this is the opening move of the "application of areas" block that continues through I.44 and I.45.

The present reconstruction preserves that synthetic character. There is no real-valued area function. Equality of content is represented by

```
HilbertScissorsEquicomplementable
```

and the constructed figure is represented by

```
hilbertParallelogramTerm
```

The final theorem therefore asserts the existence of points E, F, G such that $FECG$ is a parallelogram, its angle FEC is congruent to the prescribed angle, and the triangle term ABC is equicomplementable with the parallelogram term $FECG$.

I.42 is also a useful test of the separation between three layers that has become essential since I.35:

- **geometry**: midpoint construction, angle layoff, side-of-line information, intersection of a ray carrier with a parallel, and completion of a parallelogram;
- **content**: equal-content comparison by I.38 and the "double" relation supplied by I.41;
- **representation**: carrier identification, ray orientation, and the use of a collinearity statement rather than an invalid strict parallel statement in the boundary case $F = A$.

The classical proof is short. The formal proof is longer because one configuration fact suppressed by the classical diagram becomes decisive: the copied angle ray must actually meet the parallel through A, and the meeting point is allowed to be A itself. The final Lean file proves this rather than assuming it.

2 The Classical Configuration

Let ABC be the given triangle and let the prescribed angle be represented formally by the noncollinear triple XYZ. Euclid bisects BC at E, so $BE = EC$. He then constructs at E an angle CEF equal to the prescribed angle, choosing the new ray on the side of the line BC that contains A. Through A he draws a line parallel to EC, meeting the constructed ray at F. Through C a line parallel to EF completes the parallelogram $FECG$.

The intended final configuration is therefore

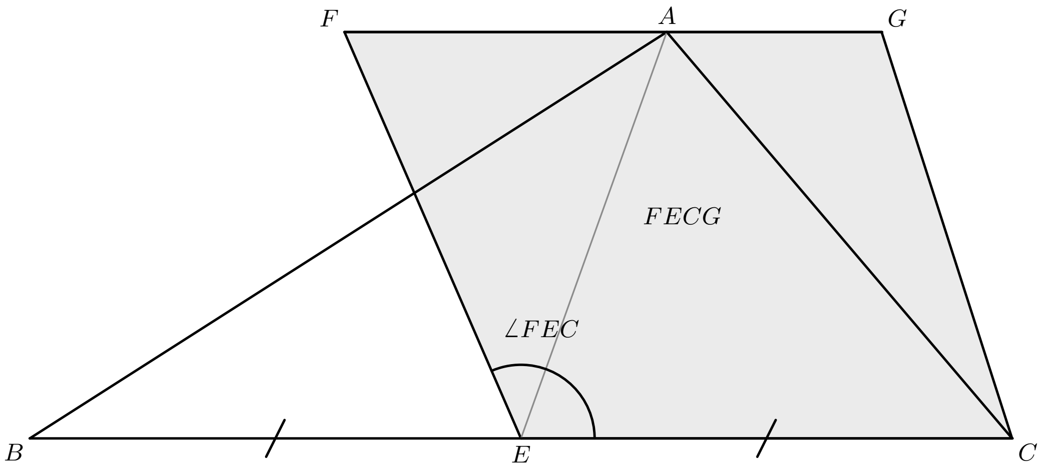
$$B - E - C, \qquad F, A, G \text{ on one upper carrier}, \qquad FG \parallel EC,$$

with $FECG$ a parallelogram and

$$\angle FEC \cong \angle XYZ.$$

The point A is not required to be strictly between F and G, and the formal construction must not exclude the legitimate case $F = A$.

Figure 1 records only the classical configuration. It uses the customary generic placement of A on the upper side of the constructed parallelogram. This placement is illustrative: the formal theorem asserts only that F, A, G are collinear and does not assert any strict order among these three points.



representative classical position

Figure 1: The classical configuration of Proposition I.42. The point E bisects BC, the prescribed angle is copied as $\angle FEC$, and the upper side of the parallelogram $FECG$ passes through A. The displayed position of A between F and G is a representative classical drawing only; no such betweenness statement is part of the formal theorem.

3 Classical Proof

Euclid's dependency graph is compact.

First bisect BC at E by I.10 and join AE. Since $BE = EC$, Proposition I.38 gives

$$\triangle ABE = \triangle AEC$$

in Euclid's equality-of-figures sense: the triangles stand on the equal bases BE and EC and between the same parallels. Hence

$$\triangle ABC = 2\triangle AEC.$$

Next construct at E the angle CEF equal to the given angle by I.23. Draw through A the line parallel to EC by I.31 and complete the parallelogram $FECG$. Proposition I.41 then gives

$$FECG = 2\triangle AEC.$$

Thus both the original triangle and the constructed parallelogram are double the same triangle, so Common Notion 1 yields

$$\triangle ABC = FECG.$$

The angle FEC has the prescribed value by construction.

The classical DAG can be displayed as

$$\begin{gathered}
E \text{ midpoint of } BC \quad \text{(I.10)} \\
\Downarrow \\
ABE = AEC \quad \text{(I.38)} \\
\Downarrow \\
ABC = 2AEC \\
\angle FEC \cong \angle XYZ \quad \text{(I.23)}, \qquad AG \parallel EC \quad \text{(I.31)} \\
\Downarrow \\
FECG \text{ parallelogram} \\
\Downarrow \\
FECG = 2AEC \quad \text{(I.41)} \\
\Downarrow \\
ABC = FECG.
\end{gathered}$$

The historical proof is therefore a forward construction. Equal content is never assumed in order to recover a metric or incidence conclusion. This will matter for the axiomatic status below.

4 Application of Areas and the Hidden Intersection

I.42 marks a change in the use of the area block. Propositions I.35–I.41 mainly compare already given figures. I.42 uses those comparison theorems as an engine for *constructing* a new figure with prescribed content and a prescribed angle. This is the first of the Book I "application of areas" constructions that will feed I.44 and I.45.

There is, however, a geometric step hidden by the ordinary diagram. After constructing the ray at E and the line through A parallel to EC, Euclid names their intersection F without separately proving that the two carriers meet. Wilkins makes the intended orientation explicit by directing the ray into the side of BC containing A. The Beeson proof trace goes further: it explicitly stops to prove that the angle ray meets the upper parallel. Its case analysis also contains the equality branch in which the intersection is A itself.

That observation changes the correct formal API. An earlier draft attempted to encode the upper-carrier condition as

```
Geo.Parallel F A E C
```

This is too strong because the project's parallel relation is strict. If $F = A$, the expression would ask the degenerate carrier AA to be parallel to EC. The final theorem instead returns

```
Collinear Geo F A G
```

for the completed parallelogram $FECG$. This says exactly what the geometry needs: A lies on the upper carrier FG, including the boundary case $F = A$.

This is not a cosmetic change of representation. It is an example of a hidden configuration condition changing the mathematically correct formal statement of an intermediate construction.

5 Proof Architecture of the Reconstruction

The final Lean proof is organized into seven conceptual stages. The local helpers are proposition-specific because they package the exact geometry of I.42 rather than a new general synthetic theory.

5.1 Step 1: midpoint and equal-content halves

The public theorem first obtains a strict midpoint E of BC. The helper

```
i42_median_bisects_area
```

proves

$$ABE \approx AEC,$$

where $\approx$ denotes `HilbertScissorsEquicomplementable`.

The proof is deliberately not a numerical same-altitude argument. It creates one remote comparison triangle DUV. A line through A and D is parallel to the base carrier. The base is extended so that the ordered points contain $B - E - C - U - V$, and a segment UV is laid off with

$$UV \cong BE \cong EC.$$

Proposition I.38 is then applied twice:

$$ABE \approx DUV, \qquad AEC \approx DUV.$$

Transitivity gives the desired midpoint-halves relation.

This reduction is a consequence of the current I.38 API: I.38 is expressed using two separated equal bases, so a common remote triangle supplies a single comparison target for both halves.

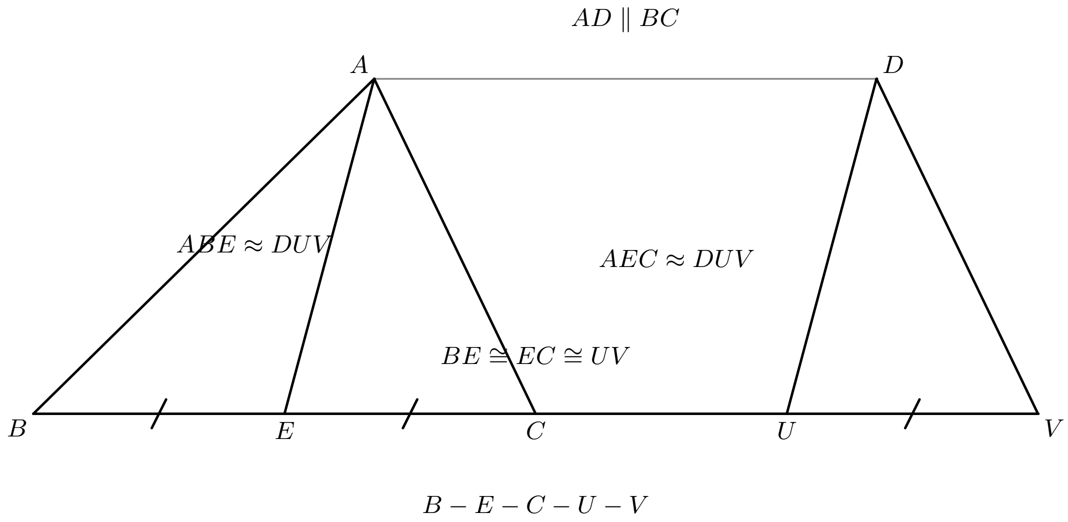


Figure 2: The auxiliary configuration used by `i42_median_bisects_area`. The base is extended in the strict order $B - E - C - U - V$; the remote triangle DUV lies on a carrier through D parallel to the original base; and $BE \cong EC \cong UV$. Proposition I.38 compares both ABE and AEC with the same triangle DUV.

5.2 Step 2: copy the angle on the correct side and draw the parallel

The helper

```
i42_angle_parallel_setup
```

constructs the carrier EC, uses Hilbert's angle-construction axiom/theorem to copy XYZ at E, and selects the side containing A. It returns a point R satisfying

$$R \text{ and } A \text{ are on the same side of } EC, \qquad \angle CER \cong \angle XYZ.$$

It also constructs a point Q such that

$$EC \parallel AQ.$$

Thus the two objects that must meet – the ray carrier ER and the upper parallel carrier AQ – are both explicit Lean data.

5.3 Step 3: prove that the two carriers meet

The helper

```
i42_angle_parallel_intersection
```

is the central Euclidean existence argument of the proposition. Suppose the lines ER and AQ were disjoint. The bridge

```
intersection_test_parallel_of_lines_disjoint
```

would convert this incidence-level disjointness into strict parallelism

$$ER \parallel AQ.$$

But $EC \parallel AQ$ as well. If the carriers ER and EC were distinct, Euclidean transitivity of parallels would imply

$$ER \parallel EC,$$

which is impossible because both contain E. If the two carriers were the same, then R would lie on the base EC, contradicting the SameSide output of angle construction. Therefore ER and AQ meet, and their intersection is named F.

This is exactly the step that the classical text suppresses.

5.4 Step 4: orient the intersection onto the correct ray

An intersection of the two *lines* is not yet enough. I.42 needs F on the chosen ray from E, not on the opposite ray. The helper

```
i42_angle_parallel_oriented_intersection
```

proves this using side-of-line data.

First, A and F lie on the same side of the base. If $F = A$, this is simply SameSide reflexivity. Otherwise the carrier AF is the carrier AQ, hence is parallel to EC, and its endpoints lie on the same side of EC. Since R and A were already constructed on the same side, transitivity gives that R and F are on the same side of EC.

Now E, R, F are collinear. They cannot satisfy $R - E - F$, because then the base line through E would separate R and F. Hence

```
HilbertSameRay Geo E R F
```

and the angle can be transported from the auxiliary ray ER to the actual intersection ray EF. The result is

$$\angle FEC \cong \angle XYZ.$$

Figure 3 separates the three distinct geometric states hidden by the usual single Euclidean sketch: construction of the two carriers, existence and orientation of their intersection, and the boundary branch in which the intersection is A itself.

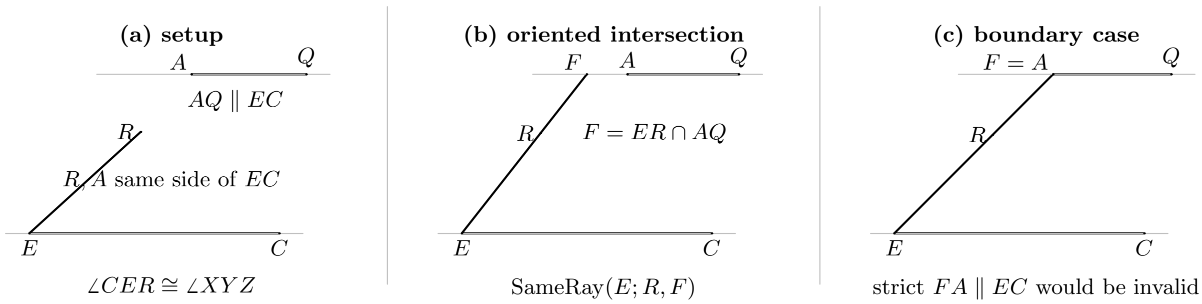


Figure 3: The formal angle/parallel intersection sequence. (a) Angle construction gives a ray carrier ER on the same side of EC as A, while $AQ \parallel EC$; no intersection witness has yet been selected. (b) The two carriers are proved to meet at F, and side-of-line information yields SameRay$(E; R, F)$; the drawing does not assert a strict order between R and F. (c) The legitimate boundary branch $F = A$ is shown literally. This branch is why the final API must not require the degenerate pair FA to be strictly parallel to EC.

5.5 Step 5: complete the parallelogram and identify its upper carrier

The helper

```
i42_construct_parallelogram
```

first proves that F is off the base EC: it lies on AQ, and AQ is parallel to EC. Thus E, F, C are noncollinear. The reusable theorem

```
hilbert_parallelogram_fourth_vertex_exists
```

then completes EFC to a parallelogram $EFGC$, which is reversed to the orientation $FECG$ used by I.42.

The upper side GF and the constructed carrier AQ are both parallel to EC and share F. Euclidean parallel uniqueness/transitivity identifies their carriers. Therefore G also lies on AQ, and the helper returns

```
Collinear Geo F A G
```

rather than the invalid strict relation $FA \parallel EC$.

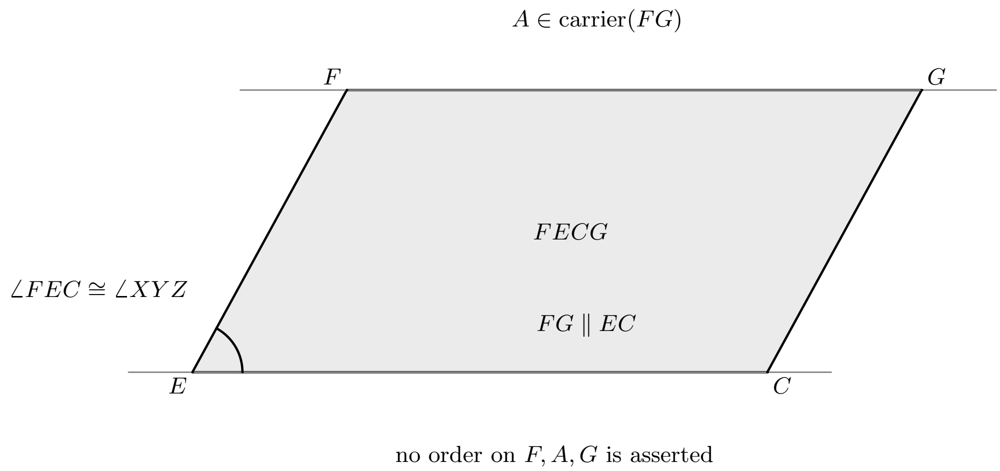


Figure 4: The geometric output of `i42_construct_parallelogram`: $FECG$ is a genuine parallelogram, $FG \parallel EC$, and the given point A lies on the carrier of FG. The figure deliberately records $A \in \text{carrier}(FG)$ without assigning any betweenness relation among F, A, G.

5.6 Step 6: use I.41 on a genuine side and transport along the carrier

The carrier formulation requires a small wrapper around I.41:

```
i42_parallelogram_double_triangle_on_carrier
```

The current public I.41 expects a strict parallel pair. Instead of using the possibly degenerate pair FA, the wrapper uses the genuine parallelogram side FG and applies I.41 with apex G:

$$\text{Par}(FECG) \approx GEC + GEC.$$

It then compares GEC with AEC. If $G = A$, this is reflexivity. Otherwise G, A, F are collinear, so the carrier GA is the same upper carrier as GF; therefore $GA \parallel EC$, and I.37 gives

$$GEC \approx AEC.$$

Using `equicomplementable_add` twice yields

$$GEC + GEC \approx AEC + AEC,$$

and transitivity gives

$$\mathrm{Par}(FECG) \approx AEC + AEC.$$

5.7 Step 7: assemble the content equality

The top-level theorem uses the scissors splitting identity

$$ABC \sim ABE + AEC,$$

where $\sim$ denotes `HilbertScissorsEq`. From Step 1 and reflexivity, `equicomplementable_add` gives

$$ABE + AEC \approx AEC + AEC.$$

Transport across the split yields

$$ABC \approx AEC + AEC.$$

Step 6 gives the same doubled triangle as the content of the constructed parallelogram. A final symmetry and transitivity step closes

$$ABC \approx \mathrm{Par}(FECG).$$

The full formal DAG is therefore

$$\begin{array}{c}
E \text{ midpoint of } BC \\
\Downarrow \\
\text{I.38 twice via } DUV \\
\Downarrow \\
ABE \approx AEC \\
\Downarrow \\
ABC \approx AEC + AEC \\
\text{angle construction } + \ EC \parallel AQ \\
\Downarrow \\
ER \cap AQ = \{F\} \quad \text{and} \quad E, R, F \text{ same ray} \\
\Downarrow \\
FECG \text{ parallelogram}, \quad F, A, G \text{ collinear} \\
\Downarrow \\
\text{I.41 at } G \quad + \quad \text{I.37 along the upper carrier} \\
\Downarrow \\
\mathrm{Par}(FECG) \approx AEC + AEC \\
\Downarrow \\
ABC \approx \mathrm{Par}(FECG).
\end{array}$$

6 Lean Formalization

The current file imports the two earlier content propositions used directly by the reconstruction:

```
import CGJteamLab.Proposition38
import CGJteamLab.Proposition41
```

The public theorem is the following. The snippet uses the project's preferred ASCII notation for documentation while preserving the exact theorem data.

```
theorem euclid_proposition_42
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C X Y Z : Geo.Point)
    (hABC : Not (Collinear Geo A B C))
    (hXYZ : Not (Collinear Geo X Y Z)) :
    exists E F G : Geo.Point,
      IsParallelogram Geo F E C G /\
      Geo.AngleCongruent F E C X Y Z /\
      HilbertScissorsEquicomplementable Geo
        (hilbertScissorsTriangle Geo A B C)
        (hilbertParallelogramTerm Geo F E C G) := by
  -- proof in Proposition42.lean
```

The six mathematically significant proposition-local helpers are:

```
i42_median_bisects_area
i42_angle_parallel_setup
i42_angle_parallel_intersection
i42_angle_parallel_oriented_intersection
i42_construct_parallelogram
i42_parallelogram_double_triangle_on_carrier
```

They are not six unrelated technical conveniences. They compress the proof into the six geometric/content transitions described above.

6.1 The midpoint-content helper

Its public shape is

```
theorem i42_median_bisects_area
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C E : Geo.Point)
    (hABC : Not (Collinear Geo A B C))
    (hE : HilbertIsMidpoint Geo E B C) :
    HilbertScissorsEquicomplementable Geo
      (hilbertScissorsTriangle Geo A B E)
      (hilbertScissorsTriangle Geo A E C) := by
  -- I.38 twice through one remote triangle DUV
```

The helper is proposition-local because its remote-triangle construction is a way of adapting the current I.38 API to the midpoint geometry of I.42. A future general same-altitude theorem might subsume it, but the present file does not introduce such a new interface theorem.

6.2 The construction helper

The decisive output is

```
theorem i42_construct_parallelogram
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A E C X Y Z : Geo.Point)
    (hAEC : Not (Collinear Geo A E C))
    (hXYZ : Not (Collinear Geo X Y Z)) :
    exists F G : Geo.Point,
      IsParallelogram Geo F E C G /\
      Geo.AngleCongruent F E C X Y Z /\
      Collinear Geo F A G := by
  -- angle ray, intersection, orientation, and fourth vertex
```

The third conjunct is the important final API decision. It records the common upper carrier without imposing strict distinctness on F and A.

6.3 The carrier form of I.41

The last local helper has the shape

```
theorem i42_parallelogram_double_triangle_on_carrier
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (F E C G A : Geo.Point)
    (hParallelogram : IsParallelogram Geo F E C G)
    (hFAG : Collinear Geo F A G) :
    HilbertScissorsEquicomplementable Geo
      (hilbertParallelogramTerm Geo F E C G)
      (hilbertScissorsTriangle Geo A E C +
       hilbertScissorsTriangle Geo A E C) := by
  -- I.41 at G, then I.37 from GEC to AEC
```

This wrapper is not a new area principle. It is a geometry/API adapter that keeps the strictness of I.41 on the genuine side FG and uses I.37 to move the triangle apex along the same upper carrier.

7 Relationship with Earlier Propositions

7.1 Proposition I.10

Classically, I.10 supplies the midpoint E of BC. The Lean theorem does not call I.10 by proposition name. It uses the developed Hilbert midpoint existence theorem directly. This is a typical Hilbert reorganization: the construction is reusable foundational infrastructure rather than a historical Book I dependency.

7.2 Propositions I.23 and I.31

Euclid uses I.23 to reproduce the given angle and I.31 to draw the parallel through A. The reconstruction again consumes the corresponding Hilbert operations directly:

```
HilbertCongruence.angle_construction
hilbert_parallel_through_point_exists
```

The formal proof then adds information absent from the classical invocation: the chosen side of the base, the line carriers, and the proof that the two constructed carriers meet.

7.3 Proposition I.37

I.37 enters only after the parallelogram has already been constructed. The carrier wrapper for I.41 first obtains the double of GEC, then uses the fact that G and A lie on the same upper carrier to obtain

$$GEC \approx AEC.$$

This is a forward use of same-base equal content.

7.4 Proposition I.38

I.38 supplies the equal-content halves cut by the median. The current I.42 proof applies I.38 twice through the auxiliary triangle DUV rather than restating a new theorem about equal altitudes. This is the main content input before the final application of I.41.

7.5 Proposition I.41

I.41 states that a parallelogram on the same base and in the same parallels is double the corresponding triangle. In I.42 the strict public API is applied with G as the apex because FG is a genuine side of the parallelogram. I.37 then transports from GEC to AEC. This avoids any need to assume $F \neq A$.

7.6 Propositions I.44 and I.45

Historically, I.42 is immediately reusable: I.44 starts by constructing a parallelogram equal to a given triangle in a prescribed angle, and I.45 builds a parallelogram equal to a general rectilinear figure by repeated application of the triangle construction. The current project follows the same broad dependency direction: Proposition I.45 imports Proposition I.42.

Cleaning the final I.42 file also exposed one dependency-hygiene issue in I.45. I.45 had been using an old proposition-local helper for adding an unchanged scissors term to an equicomplementability relation. That call was replaced by the already reusable

```
equicomplementable_add
```

from the scissors layer. No helper had to be restored to I.42, and the full project build remained clean.

8 Hilbert and Book Zero Components Used

8.1 Midpoint, betweenness, and segment construction

The midpoint data are strict Hilbert data: $B - E - C$ together with $BE \cong EC$. The proof uses betweenness incidence facts, base extensions, segment construction on a ray, and transitivity/symmetry of segment congruence to build the auxiliary equal base UV.

These operations belong to the existing Hilbert/Book Zero language. I.42 introduces no new midpoint or segment-arithmetic theorem.

8.2 Angle construction, SameSide, and SameRay

Hilbert angle construction is stronger as an API than the classical phrase "construct an equal angle": it also records the chosen side of a carrier. That SameSide witness becomes essential later. It is combined with the parallel carrier AQ to prove that the line intersection F is on the correct ray, expressed by

```
HilbertSameRay Geo E R F
```

The angle is then transported along that ray.

This is a good example of hidden order data becoming a first-class part of the formal proof rather than a diagrammatic convention.

8.3 Euclidean parallel and intersection infrastructure

The central intersection proof uses the established bridge from disjoint incidence lines to strict point-line parallelism, together with Euclidean transitivity of distinct parallels and the fact that parallel carriers cannot share a point. Carrier identification later uses the same Euclidean uniqueness principle.

Thus the parallel axiom enters I.42 not merely because a parallel line is drawn, but because the proof must control uniqueness and intersection of the constructed carriers.

8.4 Parallelogram infrastructure

Once E, F, C are shown noncollinear, the theorem

```
hilbert_parallelogram_fourth_vertex_exists
```

constructs the fourth vertex. The representation is normalized by

```
ParallelogramReverse
```

and the upper side is identified with the already constructed carrier through A. No new parallelogram axiom is introduced.

8.5 Scissors content infrastructure

The content layer uses only previously established operations:

```
equicomplementable_add
equicomplementable_refl
equicomplementable_symm
equicomplementable_trans
equicomplementable_transport
```

The triangle split itself is a `HilbertScissorsEq` identity. At no point is a real-valued area, scalar multiplication, subtraction, or content order introduced.

9 Source Audit

9.1 Euclid, Heath, and Joyce

The classical text gives the exact short DAG used as the historical reference: bisect BC, construct the prescribed angle at E, draw the parallel through A, complete the parallelogram, use I.38 for the two half-triangles, and use I.41 for the parallelogram. Joyce also identifies I.42 as the beginning of the application-of-areas phase of Book I.

This source controls the statement and the classical dependency graph. It does not by itself resolve the hidden existence proof for the intersection point F.

9.2 Wilkins

Wilkins makes one orientation choice explicit that is important for the formal reconstruction: the newly constructed ray from E is directed into the side of the line BC containing A. His account then describes the parallel through A as meeting that ray at F and completes the parallelogram. This is the correct classical configuration control for the SameSide/SameRay argument in Lean.

9.3 Beeson

The Beeson formal proof corpus is especially useful as a gap audit here. Its I.42 trace explicitly interrupts the Euclidean construction to prove that the angle ray meets the upper parallel, noting that this meeting is omitted in Euclid's prose. More importantly, the trace includes a branch in which the meeting point is A itself.

The final Lean architecture is not copied from the Tarski-style Beeson proof, but these observations directly validate two design decisions: intersection must be proved, and the intermediate API must allow $F = A$.

9.4 Hilbert

Hilbert's Chapter IV remains the semantic control for the content block. The distinction between congruent-piece equivalence and equicomplementability is maintained throughout the project. In I.42 the final proof does not call Hilbert's Theorems 45–46 directly; instead their content architecture is already embodied in the previously proved I.37–I.41 layer.

For the geometric construction, the Hilbert organization is even more direct: angle layoff, side-of-line data, line incidence, and the Euclidean uniqueness of parallels are reusable operations rather than historical Book I propositions.

9.5 Hartshorne

Hartshorne's Section 22 is the main semantic check on the forward content argument. He distinguishes the results that require the extra strictness property (Z) from those that remain valid for equal content. The remaining Book I results I.41–I.45 are in the latter class.

That classification matches the actual Lean DAG. I.42 never starts from an equal-content hypothesis and concludes a metric or incidence fact. It only constructs geometry and then combines forward equal-content statements. Accordingly no cancellation, positivity, whole-greater-than-part principle, or property (Z) is needed.

Hartshorne also places I.42–I.45 in the "algebra of areas" or application of areas: synthetic content is manipulated constructively without assigning a numerical area value to each figure.

9.6 Borsuk–Szmielew and Greenberg

Borsuk–Szmielew are strongest here as a control source for the underlying synthetic API. Their organization emphasizes angle layoff on a chosen side, midpoint data with positional information, and – after the Euclidean axiom – parallel uniqueness, parallel transitivity, and parallelogram theory. These are precisely the kinds of components used by the geometric half of I.42.

They do not supply the equal-content semantics of the I.35–I.45 block. As in the preceding documentation, that role remains with Hilbert and Hartshorne. Greenberg likewise serves as a control on the Euclidean parallel boundary and on the distinction between synthetic content and numerical area rather than as the source of an I.42-specific proof.

10 Formalization Notes

10.1 Geometry

The formal geometry contains substantially more data than the classical picture displays:

- strict noncollinearity for the input triangle and prescribed angle;
- strict midpoint betweenness $B - E - C$;
- a concrete carrier line for EC;
- the fact that the copied angle point R and A lie on the same side of that carrier;
- a concrete carrier AQ parallel to EC;
- a proof that the ray carrier ER meets AQ;
- a proof that the resulting intersection F lies on the selected ray;
- noncollinearity of E, F, C before the fourth vertex can be constructed;
- identification of the completed upper side FG with the carrier through A.

No Pasch step is introduced locally in the top-level theorem, but the proof uses the developed side-of-line and intersection theory whose lower layers are built from Hilbert order/separation infrastructure.

10.2 Content

The content argument is entirely forward:

$$ABE \approx AEC \Longrightarrow ABC \approx AEC + AEC,$$

and

$$\mathrm{Par}(FECG) \approx GEC + GEC \Longrightarrow \mathrm{Par}(FECG) \approx AEC + AEC.$$

Transitivity then compares the original triangle with the constructed parallelogram.

There is no inverse "halving" theorem. There is no cancellation of a common summand and no deduction from equal content to segment or angle congruence. Thus I.42 stays on the safe forward side of the content semantics established since I.35.

10.3 Representation

Three representational choices deserve explicit mention.

First, the angle-construction helper initially returns the angle in the order CER and the final theorem wants FEC. SameRay transport and the symmetry of angle representation normalize this without changing the geometry.

Second, the parallelogram fourth-vertex theorem naturally returns $EFGC$, while the Euclid statement is expressed as $FECG$. The theorem `ParallelogramReverse` performs this normalization.

Third, and most importantly, the upper-carrier datum is

```
Collinear Geo F A G
```

rather than a strict parallel using the pair FA. This is the correct representation because the pair may degenerate when $F = A$ even though the upper carrier itself is perfectly nondegenerate through F and G.

10.4 Axiomatic status

The public theorem assumes

```
HilbertEuclideanPlane Geo
```

and is therefore Euclidean in the current hierarchy.

The Euclidean parallel axiom is used essentially in the developed theory of parallel uniqueness/transitivity. The most visible direct use in I.42 is the proof that the angle-ray carrier and the parallel through A cannot remain disjoint: two distinct carriers parallel to the same carrier are forced into the Euclidean parallel relation, which contradicts their shared point when appropriate. The same uniqueness theory identifies the final upper carrier.

Angle construction and segment construction are congruence/order operations; they do not require a continuity axiom in this proof. No circle intersection or completeness principle is introduced. Proposition I.42 adds no new geometric axiom and no new content axiom.

11 Dependency Summary

The actual formal dependency chain is

$$\begin{gathered}
\text{Hilbert incidence, order, congruence, and Euclidean parallel axioms}\\
\Downarrow\\
\text{HilbertInterface / Book Zero line, ray, side-of-line, midpoint theory}\\
\Downarrow\\
\text{angle construction + parallel-through-point + intersection theory}\\
\Downarrow\\
\text{Proposition I.38 twice}\\
\Downarrow\\
\texttt{i42_median_bisects_area}\\
\Downarrow\\
ABC \approx AEC + AEC\\
\texttt{i42_angle_parallel_setup}\\
\Downarrow\\
\texttt{i42_angle_parallel_intersection}\\
\Downarrow\\
\texttt{i42_angle_parallel_oriented_intersection}\\
\Downarrow\\
\texttt{i42_construct_parallelogram}\\
\Downarrow\\
\text{I.41 + I.37 + scissors addition}\\
\Downarrow\\
\texttt{i42_parallelogram_double_triangle_on_carrier}\\
\Downarrow\\
\texttt{euclid_proposition_42}.
\end{gathered}$$

The historical and formal DAGs have the same content skeleton but different geometric packaging:

$$\begin{aligned}
\text{Euclid} &: \quad \text{I.10} \to \text{I.38} \to \text{I.23/I.31} \to \text{I.41} \to \text{I.42},\\
\text{Lean} &: \quad \text{Hilbert midpoint} \to \text{I.38 twice} \to \text{angle/parallel intersection} \to \text{I.41+I.37} \to \text{I.42}.
\end{aligned}$$

New geometric axiom in I.42:	no
New content axiom in I.42:	no
New proposition-local helpers:	yes — six
New reusable Interface/Scissors theorem:	no
New Book Zero theorem:	no
Property (Z) / content cancellation:	no
New numerical area semantics:	no
Continuity axiom introduced:	no
Boundary case $F = A$ supported:	yes
Public theorem compiles:	yes
Full `lake build`:	clean
`sorry` in `Proposition42.lean`:	no

Diagrammatic audit. The revised diagram layer deliberately separates the classical and formal configurations. Figure 1 is the historical construction and therefore permits a generic visual placement of A on the upper side. Figure 2 records the genuinely new auxiliary geometry of the midpoint-content helper. Figure 3 records the three successive angle/parallel states, including the essential boundary case $F = A$. Figure 4 records the actual output of the formal parallelogram construction without inventing an order on F, A, G.

No additional figure is introduced for the exact split $ABC \sim ABE + AEC$, for the applications of I.41 and I.37, or for the final symmetry, addition, transport, and transitivity steps. Those operations change the scissors or logical interpretation of geometry already displayed; they do not introduce a new geometric configuration.

12 Conclusion

Proposition I.42 is a construction theorem built on the content theory of I.35–I.41, but the main formal difficulty is geometric rather than algebraic. The equal-content calculation itself is simple once the right configuration has been produced: the midpoint divides the original triangle into two equal content pieces, I.41 makes the new parallelogram double a triangle on its base, and I.37 moves that triangle along the common upper carrier.

What the formal reconstruction adds is the configuration data that the classical diagram suppresses. The copied angle ray is placed on the correct side, its carrier is proved to meet the upper parallel, the intersection is proved to lie on the correct ray, and the completed upper side is identified with the carrier through A. The boundary case $F = A$ is not discarded; it is the reason the final construction returns collinearity on the upper carrier instead of a strict parallel relation involving FA.

The proposition therefore illustrates the mature architecture of the Book I reconstruction. Geometry constructs and orients the carriers; the scissors layer carries equal content; representation lemmas normalize the same figure without changing its meaning. No new axiom, numerical area, cancellation principle, or continuity assumption is needed. The final full project build is clean, and I.42 is available as the construction input required by the later application-of-areas propositions.